

ABDELRAHMAN ELDERAA

Data Scientist | Machine Learning | NLP

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with 1+ year of hands-on experience building end-to-end machine learning pipelines across classification, NLP, and recommendation systems. Proven ability to handle real-world challenges including severe class imbalance (SMOTE on 1.85M transactions), large-scale NLP preprocessing (TF-IDF + Lemmatization on 10K songs), and advanced feature engineering. Additional expertise in Computer Vision (YOLOv8, UNet++, Medical Imaging) and LLM/RAG systems using LangChain, FAISS, and Google Gemini. Currently pursuing MSc in Data Science at Cairo University.

EDUCATION

M.Sc. Data Science | *Cairo University* 2026 – Present

- First year — coursework phase covering advanced ML, statistical modeling, and data engineering.

B.Sc. Computer Science | *Modern Academy, Cairo* Oct 2020 – Jun 2024

- GPA: 3.57 / 4.0 — Top academic performance.
- ECPC (Egypt Collegiate Programming Contest) participant.

PROFESSIONAL EXPERIENCE

AI Engineer — Computer Vision | *Balance Code Company* Sep 2025 – Present

- Developed and optimized deep learning models for real-time image/video analysis using YOLO and Attention-based CNNs.
- Built end-to-end CV pipelines: data ingestion, preprocessing, training, evaluation, and Streamlit deployment.

AI Instructor — ML & Computer Vision | *Mindset Company* Feb 2025 – Sep 2025

- Delivered hands-on workshops on image segmentation, object detection, and regression using Python and TensorFlow.
- Mentored 30+ students through CV and ML capstone projects; designed project-based curriculum.

Supervisor & Program Coordinator | *Mindset Company* Jul 2024 – Jan 2025

- Managed AI course delivery, tracking student progress and providing structured feedback on Python and ML fundamentals.

Teaching Assistant — Computer Science | *Modern Academy* Aug 2024 – Aug 2025

- Conducted tutorials on algorithms, data structures, and programming for undergraduate students.

KEY PROJECTS

Credit Card Fraud Detection — *Python · XGBoost · SMOTE · Scikit-learn · Streamlit*

- Built an end-to-end fraud detection pipeline on 1.85M real transactions with extreme class imbalance (0.52% fraud); applied SMOTE after train-test split to prevent data leakage, balancing ~1.47M samples per class.
- Engineered 6 new features (log_amt, distance, age, hour, day_of_week, month) improving model signal; trained XGBoost with L1/L2 regularization achieving 95.5% ROC-AUC and 95% fraud recall.
- Deployed interactive Streamlit dashboard with full evaluation: Confusion Matrix, Classification Report, ROC Curve, and Feature Importance visualization.

Music Recommendation System — *Python · TF-IDF · Cosine Similarity · NLTK · Streamlit*

- Built a content-based recommendation engine on 10,000 Spotify songs using full NLP pipeline: Lowercasing, Punctuation Removal, Tokenization, Stopword Removal, and WordNet Lemmatization.
- Vectorized lyrics using TF-IDF (5,000 features) with on-demand Cosine Similarity computation to avoid MemoryError on large matrices; deployed as interactive Streamlit app with similarity progress bars.

News Research Tool & RAG Chatbot — *LangChain · FAISS · Llama 3 · Gemini · Streamlit*

- Built two production RAG Q&A systems: news research tool using LangChain + Llama-3.3-70B + FAISS with automated URL scraping and chunking; knowledge-base chatbot using Google Gemini + HuggingFace embeddings.
- Both deployed via Streamlit with hallucination-controlled prompting; demonstrated end-to-end LLM pipeline from ingestion to inference.

Loan Status Prediction — *Python · SVM · Scikit-learn · EDA*

- Predicted loan approval on 614 applications using SVM (linear kernel); applied Label Encoding for 6 categorical features and stratified 90/10 train-test split achieving 83.3% test accuracy.
- Performed full EDA with visualizations: approval rates by education, marital status, property area, and dependents.

Coronary Artery FFR Estimation System — *Python · PyTorch · UNet++ · OpenCV*

- Built deep learning pipeline using UNet++ (VGG16-BN encoder) to segment coronary arteries from angiography videos; trained on 9,867 pairs with custom loss (cIDice $\times 0.7$ + Focal Tversky $\times 0.3$) achieving Dice 87.50% / IoU 79.29%.
- Implemented QCA to estimate FFR using Wibmer 2018 formula, classifying results into 4 severity levels with annotated video overlay and per-vessel clinical CSV reports.

CERTIFICATIONS & TRAINING

- Huawei HCIA-AI V3.5 — NTI (Jun 2025 – Jul 2025)
- Artificial Intelligence Diploma — AMIT Learn (Jul 2023 – Mar 2024) | Python, ML, DL, NLP, CV
- AI & Machine Learning Internship — Instant (Jun 2023 – Aug 2023)

CORE TECHNICAL SKILLS

Data Science & ML	Scikit-learn, XGBoost, SVM, Logistic Regression, Random Forest, SMOTE
NLP & Text Mining	TF-IDF, Cosine Similarity, NLTK, spaCy, Lemmatization, Tokenization
LLMs & RAG	LangChain, FAISS, Pinecone, Google Gemini, Llama 3, HuggingFace, Cohere
Deep Learning	TensorFlow, Keras, PyTorch, CNNs, U-Net, UNet++, YOLOv8, Transfer Learning
Data & Visualization	NumPy, Pandas, Matplotlib, Seaborn, Power BI
Deployment & Tools	Streamlit, FastAPI, Flask, Git, GitHub, Jupyter, Google Colab, Roboflow
Languages	Python (primary), SQL, C++, Java

LANGUAGES

- Arabic — Native proficiency
- English — Intermediate (reading, writing, technical communication)